

Lab 2: Agile Backlog Creation & Sprint Simulation in Jira

Scenario: Problem Statement #18 - Clinical Trial Subject Recruitment Engine (Healthcare & Telemedicine)

Actors: Trial Candidate, Clinical Researcher

Jira project: My Scrum Space (key SCRUM), Scrum board

Totals: 4 epics, 13 user stories, 64 story points, 2 sprints (30 + 34 points), all stories Done

1. Epics

Key	Epic	Description	Priority	Stories	Points
SCRUM-1	Candidate Registration & Eligibility	Register trial candidates and complete eligibility screening against the trial inclusion and exclusion criteria.	Highest	4	19
SCRUM-2	Informed Consent Management	Capture and track informed-consent milestones: document viewing, e-signature and witness confirmation.	High	4	16
SCRUM-3	Trial Visit & Compliance Tracking	Schedule trial visits, record attendance and actual dates, and track protocol deviations.	Medium	3	13
SCRUM-4	Audit & Compliance Reporting	Generate audit-trail reports with tamper-evident audit information (21 CFR Part 11).	Medium	2	16
	Total			13	64

2. User Stories, Priority and Story Points (Fibonacci scale)

Key	User story	Epic	Priority	Points	Sprint	Status
SCRUM-5	Story 1.1: Register Candidate Profile	SCRUM-1	High	3	Sprint 1	Done
SCRUM-7	Story 1.2: Complete Eligibility Screening Survey	SCRUM-1	High	5	Sprint 1	Done
SCRUM-8	Story 1.3: Evaluate Eligibility Against Trial Criteria	SCRUM-1	Highest	8	Sprint 1	Done
SCRUM-9	Story 1.4: View Failed Criteria for Ineligible Candidates	SCRUM-1	High	3	Sprint 1	Done
SCRUM-10	Story 2.1: Review Informed Consent Document	SCRUM-2	High	3	Sprint 1	Done
SCRUM-11	Story 2.2: E-Sign Informed Consent	SCRUM-2	Highest	5	Sprint 1	Done
SCRUM-12	Story 2.3: Confirm Witness Signature	SCRUM-2	Medium	3	Sprint 1	Done
SCRUM-13	Story 2.4: Track Consent Milestones	SCRUM-2	Medium	5	Sprint 2	Done
SCRUM-14	Story 3.1: Schedule Trial Visits	SCRUM-3	High	5	Sprint 2	Done
SCRUM-15	Story 3.2: Record Visit Attendance and Actual Dates	SCRUM-3	Medium	3	Sprint 2	Done
SCRUM-16	Story 3.3: Flag Protocol Deviations	SCRUM-3	Medium	5	Sprint 2	Done
SCRUM-17	Story 4.1: Generate Audit Trail Report	SCRUM-4	Medium	8	Sprint 2	Done
SCRUM-18	Story 4.2: Maintain Tamper-Evident Audit Log	SCRUM-4	High	8	Sprint 2	Done

Points used: 3, 5 and 8 from the Fibonacci sequence (0, 1, 2, 3, 5, 8, 13...). 8 points was reserved for stories with the most logic and uncertainty (eligibility rule evaluation, audit report, cryptographic audit log); simple capture or view screens got 3.

3. Sprint Plan and Sprint Board

Sprint	Goal	Dates	Stories	Points	To Do / In Progress / Done
Sprint 1	Deliver candidate registration, eligibility screening and the core informed-consent flow	28 Sep 2026 - 05 Oct 2026 (1 week)	7	30	0 / 0 / 7
Sprint 2	Deliver consent tracking, trial visit scheduling and compliance, and audit reporting	28 Sep 2026 - 05 Oct 2026 (1 week)	6	34	0 / 0 / 6

Sprint 1 board (final state)

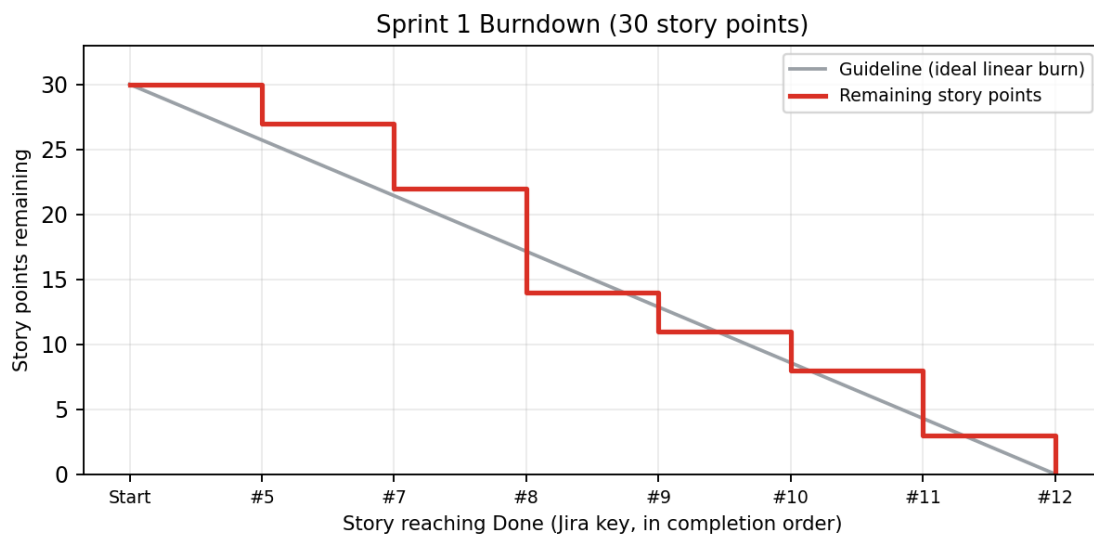
To Do	In Progress	Done
(empty)	(empty)	SCRUM-5 - Register Candidate Profile (3 pts) SCRUM-7 - Complete Eligibility Screening Survey (5 pts) SCRUM-8 - Evaluate Eligibility Against Trial Criteria (8 pts) SCRUM-9 - View Failed Criteria for Ineligible Candidates (3 pts) SCRUM-10 - Review Informed Consent Document (3 pts) SCRUM-11 - E-Sign Informed Consent (5 pts) SCRUM-12 - Confirm Witness Signature (3 pts)

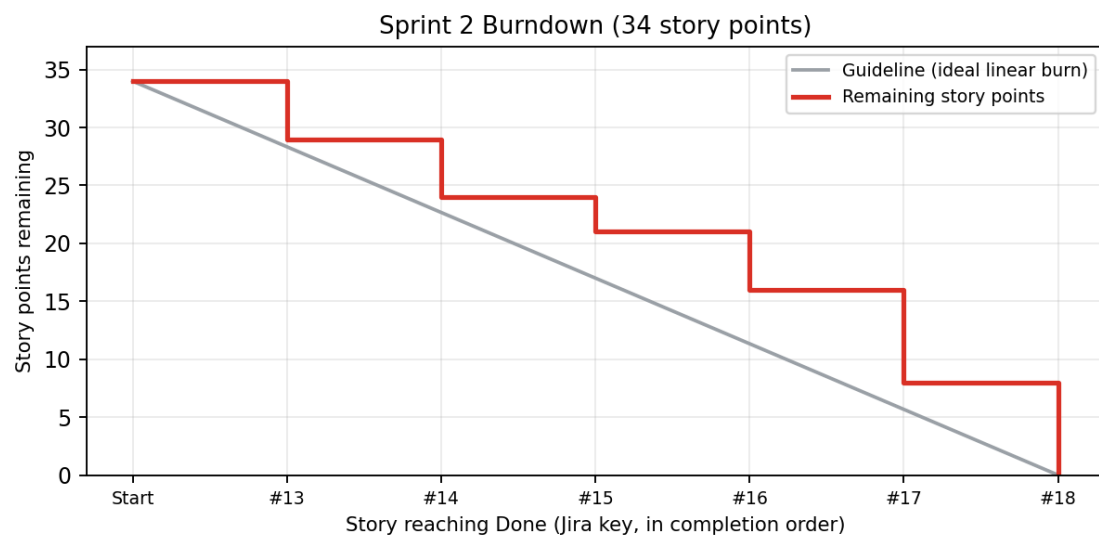
Sprint 2 board (final state)

To Do	In Progress	Done
(empty)	(empty)	SCRUM-13 - Track Consent Milestones (5 pts) SCRUM-14 - Schedule Trial Visits (5 pts) SCRUM-15 - Record Visit Attendance and Actual Dates (3 pts) SCRUM-16 - Flag Protocol Deviations (5 pts) SCRUM-17 - Generate Audit Trail Report (8 pts) SCRUM-18 - Maintain Tamper-Evident Audit Log (8 pts)

Every story was moved To Do to In Progress to Done in Jira and each sprint was completed with 0 unfinished items.

4. Burndown Charts





Charts are regenerated from the Jira story-point and completion data (remaining points after each story reached Done). Jira's own Reports > Burndown Chart for Sprint 1 and Sprint 2 shows the same data.

5. Reflection Questions

1. Did your estimations reflect the actual effort?

Only partly, and this needs an honest caveat: the sprints were simulated, so no real development effort was spent and the estimates could not be checked against actual hours. What the estimates did capture was relative size. The eligibility evaluation engine (8), the audit report (8) and the cryptographic audit log (8) carry rule logic, data integrity and compliance risk, while profile registration, viewing failed criteria and document review (3 each) are mostly forms and read views. In a real sprint I expect the 8-point items to be the ones that run over, and I would split them into smaller stories (for example, separate criteria configuration from criteria evaluation).

2. Was your backlog well-prioritized?

Mostly yes. Eligibility evaluation (FR-001) and e-signature of consent are Highest because every later step depends on them: no subject can be enrolled or scheduled without an eligibility decision and valid consent. Registration and screening were placed before them because they feed those stories. One weakness: the tamper-evident audit log (NFR-001, High) sits in Sprint 2, although 21 CFR Part 11 audit trails are safest when built in before records are created. On a real project I would pull a thin version of it into Sprint 1.

3. How did your simulated sprint align with your plan?

Both sprints matched the plan exactly: Sprint 1 committed 7 stories / 30 points and Sprint 2 committed 6 stories / 34 points, and all 13 stories (64 points) reached Done with nothing carried over and no scope added mid-sprint. That perfect result is a product of the simulation, since the stories were moved through the board in minutes. Real sprints normally show some spill-over, so full completion here should not be read as proof that the plan was realistic.

4. What insights did the burndown chart give about your team's capacity?

The charts show a stepped line: each story's points drop off only when the story is Done, so large stories (the 8-point eligibility engine in Sprint 1, the two 8-point audit stories in Sprint 2) cause large, lumpy drops, a sign that they should be broken down for smoother tracking. The planned capacity was 30 and 34 points per sprint, so a working velocity of about 30 to 35 points is a reasonable starting assumption. Because the whole simulated sprint was compressed into minutes, the red line sits far below the guideline almost immediately; in a real sprint that pattern would suggest the team under-committed, so the velocity figure should be re-measured with real work before it is used for planning.